

Total No. of Questions : 8]

SEAT No. :

PC2349

[Total No. of Pages : 2

[6354]-465

B.E. (Civil)

AIR POLLUTION AND CONTROL

(2019 Pattern) (Semester - VII) (Elective - IV) (401004A)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Draw neat figures wherever necessary.
- 4) Assume suitable data if necessary.
- 5) Use of scientific calculators is allowed.

- Q1)** a) Enlist various methods of sampling particulate matter and explain any one in detail. **[6]**
- b) Explain the purpose of ambient air sampling. **[6]**
- c) Write a short note on High volume sampler with sketch. **[6]**

OR

- Q2)** a) Write in a tabular form National ambient air quality standard (NAAQS) specified by central pollution control board (CPCB). **[6]**
- b) Explain Stack monitoring in details with a neat sketch. **[6]**
- c) Convert $80 \mu\text{g}/\text{m}^3$ of SO_2 in ppm. Assume temperature 25°C and pressure at 103.193 kPa. **[6]**

- Q3)** a) State the Basic equation of Emission estimation. **[6]**
- b) Explain types of air quality models. **[6]**
- c) Deliberate the strengths and limitations of AERMOD model USEPA. **[6]**

OR

- Q4)** a) Write a short note on air quality modelling. **[6]**
- b) Explain Emission inventory framework developed by CPCB. **[6]**
- c) Compare the physical, statistical and deterministic air quality models. **[6]**

P.T.O.

Q5) a) Calculate the minimum size of the particle that will be removed with 100% efficiency from a settling chamber for the following data: **[6]**

- i) Horizontal velocity – 0.25 m/s
- ii) Particle Density - 2500 kg/m³
- iii) Chamber: Length - 7.5 m, height - 1.5 m
- iv) Viscosity of air - 1.9×10^{-5} kg/m s. (v) Gas Density – 1.0888 kg/m³

The settling chamber will be operated under quiescent conditions.

- b) Write a note on electrostatic precipitator on mechanism, working principle & applications. **[6]**
- c) Explain the measures for controlling the emission from mobile sources. **[5]**

OR

Q6) a) State the working principle, advantages and disadvantages of fabric filter as a particulate control equipment. **[6]**

- b) Describe the factors responsible for selection of particulate control equipment. **[6]**
- c) What is carbon sequestration? Enlist the types, explain any one. **[5]**

Q7) a) Enlist methods for odour pollution control. Explain any one in detail. **[5]**

- b) Explain sick building syndrome with causes and its preventive measures. **[6]**
- c) What is active and passive sampler? Explain with advantages and disadvantages. **[6]**

OR

Q8) a) What are the causes and effects of indoor air pollution? **[5]**

- b) Explain use of plants for indoor air quality improvement. **[6]**
- c) Enumerate the odorous materials with respect to following industries **[6]**
 - i) Fertilizer
 - ii) Petroleum
 - iii) Chemical

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